



PULSE

Pulse, Explained

From “what is a risk score?” to a grounded, agentic clinical-decision-support platform — a plain-English tour of aortic & endovascular surgery, from referral to recovery.

DECISION SUPPORT

GROUNDED AI

ZERO PHI · SYNTHETIC

AGENTIC

```
pulse@ward:~$ ./read --audience=clinician,engineer --depth=full
```

Clinical decision support for aortic surgery
Next.js · FastAPI · Quiver RAG · deterministic
engine

Authored for clinicians &
engineers
github.com/achref-sova/pulse

How to read this guide & what's inside

Pulse is an educational clinical-decision-support demonstrator for aortic and endovascular surgery. This guide explains what it does and how it is built — for two readers at once. Wherever a *clinician* wants to know *what* a feature decides, a paragraph explains it in plain terms. Wherever an *engineer* wants to know *how*, an  Under the hood aside or a diagram shows the mechanism. Read straight through, or jump to a chapter.

READ THIS FIRST

Pulse runs entirely on **synthetic** patient data. Nothing in it is real, and nothing in it is medical advice. It exists to demonstrate a way of building clinical software where **the AI never invents a number** — every score is computed by tested code and every claim is traceable to a source.

- 1 From Referral to Recovery** p4
What Pulse is, the wedge, and the one rule that shapes everything.
- 2 The Clinical Journey** p6
Intake → risk → plan → monitor — the phases of a case, and where Pulse helps.
- 3 The Architecture, Top to Bottom** p7
The whole system on one page: web, API, intelligence, and data planes.
- 4 The Deterministic Engine** p9
Seven tested calculators and an IFU-fit model — why the maths lives in code.
- 5 The Agentic Brain** p11
Tools, a router, memory, grounding, and streaming — the copilot that reasons.
- 6 Retrieval and Knowledge** p13
How Pulse grounds answers in guidelines with the Quiver vector database.

7	Privacy by Design	p14
	<i>Zero-PHI, synthetic data, tenancy, and honest non-goals.</i>	
8	The Decisions Behind the Code	p15
	<i>The architectural choices — and the restraint — recorded as ADRs.</i>	
9	Beyond the App	p16
	<i>The MCP server, the numbers, and how an external agent drives Pulse.</i>	
A	Glossary for the newcomer	p17
B	References and further reading	p18

From Referral to Recovery

A patient with an aortic aneurysm travels a long road — referral, imaging, risk assessment, a device decision, surgery, and recovery. Pulse is a decision-support companion for that whole road.

An **abdominal aortic aneurysm** is a focal dilation of the aorta. Left alone, a large one can rupture — an event that is frequently fatal. Managing it is a sequence of judgements: is it big enough to fix? Is this patient fit for the operation? Which stent-graft fits this anatomy? How is the patient trending after surgery? Each judgement leans on numbers — risk indices, anatomical measurements, vital-sign scores — and on evidence from published guidelines.

Pulse is a rebuild of the author's first project, *CardioSurg-AI-Assistant* (a Streamlit + ChromaDB prototype), onto a modern, agentic stack. The prototype proved the idea; Pulse makes it production-shaped: typed, tested, multi-tenant, observable, and — crucially — **grounded**.

1.1 The wedge: grounded, not generative

Plenty of tools will now write a clinical-sounding paragraph. Very few will guarantee that the RCRI score in that paragraph was *computed*, from *this* patient's chart, by *tested* code — and that the guideline it cites actually exists. That guarantee is Pulse's wedge.

1 Deterministic where it must be

Every clinical score comes from a small, tested calculator — never from a language model. The model may *read* a value off the chart, but the arithmetic is code. *A score is earned, not guessed.*

2 Agentic where it helps

A tool-calling agent decides *which* calculator to run, fetches the record, matches devices, and searches the literature — then explains its reasoning, step by step, with citations. *The AI orchestrates; it does not fabricate.*

3 Private by construction

The entire demo runs on synthetic patients. There is no PHI to leak, and the architecture — RLS, tenancy, service tokens — is built as if there were. *Own the pattern before you own the data.*

THE ONE RULE

The language model never invents a clinical score. It delegates to the deterministic engine, and it grounds every clinical claim in a tool result or a cited guideline. Everything else in Pulse follows from that rule.

△ Educational demo on synthetic data – not for clinical use; not medical advice.

The Clinical Journey

A case moves through phases. Pulse attaches a decision-support surface to each one — and mirrors the same shape Uncvr uses for surgical video: capture, analyse, report.

Every patient in Pulse carries a *phase* — pre-operative, intra-operative, or post-operative — and a *planned intervention*. The dashboard, the roster, and the copilot all read that context, so the tool always knows where in the journey a patient stands.



Figure 2.1 — The referral-to-recovery journey, and where Pulse's surfaces attach.

The copilot sits across all four phases. Ask it “is this patient suitable for EVAR?” and it will fetch the record, run the fit engine, and answer with the per-criterion verdict — the same numbers the Plan surface shows, because they come from the same code.

The Architecture, Top to Bottom

Two languages, by design: TypeScript for the product surface, Python for the intelligence and data plane. Each does what it is best at.

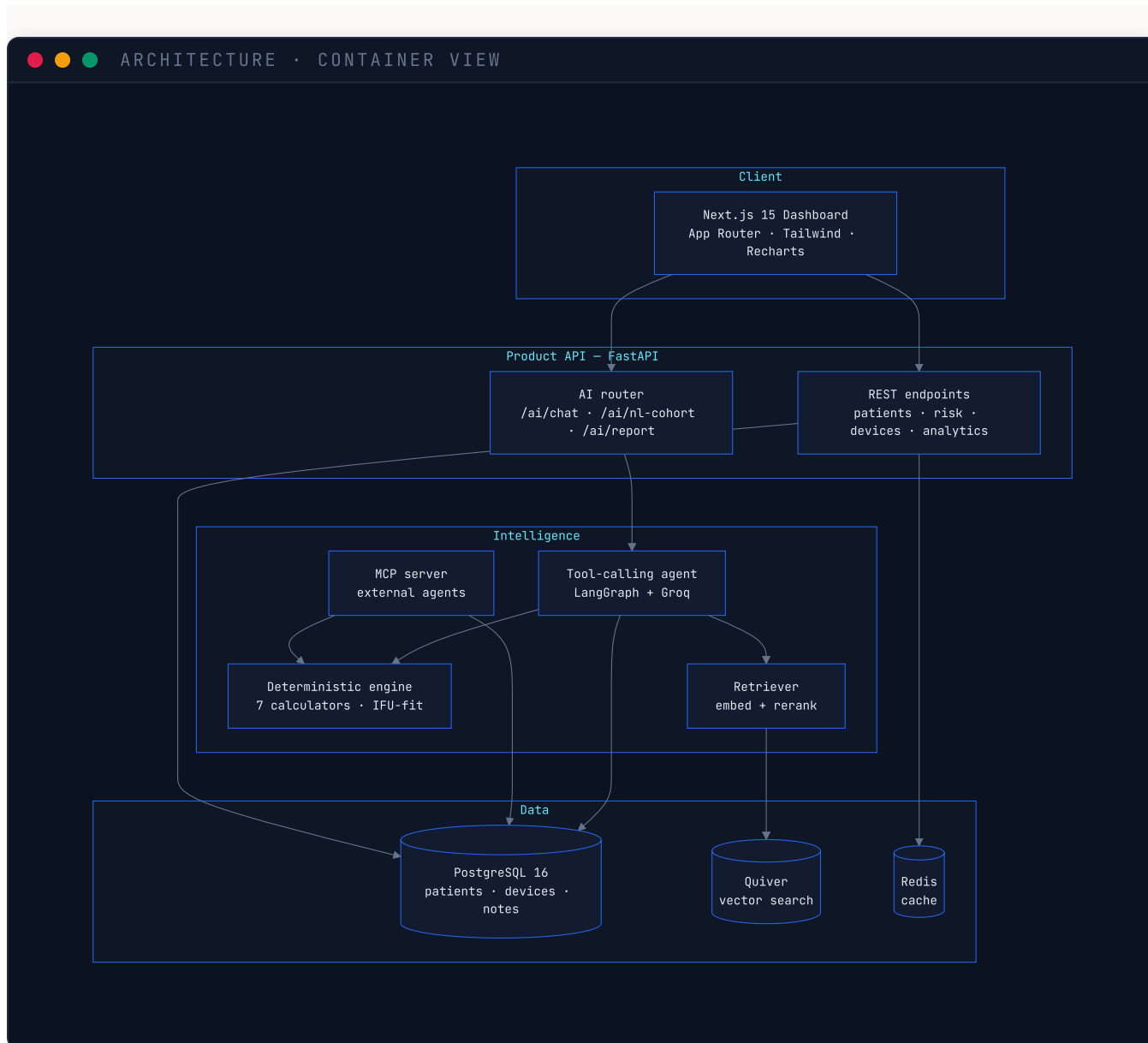


Figure 3.1 – The container view. The agent and the MCP server share one deterministic engine.

3.1 Why polyglot

The dashboard is a [Next.js 15](#) app — App Router, server components, Tailwind with a Clinical Cobalt token system that works in light and dark. The product API and the intelligence plane are [FastAPI](#) (Python 3.12, managed with [uv](#)): async SQLAlchemy over [PostgreSQL 16](#), the clinical calculators, the LangGraph agent, and the retriever. Vectors live in [Quiver](#) — the author’s own Rust vector database — and [Redis](#) handles cache.

🔍 UNDER THE HOOD

The web tier never touches Postgres directly; it goes through the typed FastAPI gateway. That keeps one authorization boundary and one place to reason about tenancy. The agent’s tools open their own request-scoped database session, bound per turn — so a tool call inside the graph reads the same session the request opened.

The Deterministic Engine

Seven clinical calculators and an anatomical fit model. Small, pure, tested — and matched to their published references.

This is the ground everything else stands on. A clinical score is not a matter of opinion: it is a defined function of defined inputs. Pulse implements each one as a small pure function with a typed input and a typed result, and covers it with unit tests against worked examples from the source publication.

The clinical calculators — each a pure, tested function

SCORE	WHAT IT ESTIMATES	REFERENCE
RCRI	Peri-operative cardiac risk (Lee index)	Lee <i>et al.</i> , Circulation 1999
NEWS2	Early-warning deterioration score	RCP, 2017
GAS	Glasgow Aneurysm Score, AAA operative risk	Samy <i>et al.</i> , 1994
EuroSCORE II	Cardiac-surgery mortality	Nashef <i>et al.</i> , 2012
CHA ₂ DS ₂ -VASc	Stroke risk in atrial fibrillation	Lip <i>et al.</i> , 2010
HAS-BLED	Major-bleeding risk on anticoagulation	Pisters <i>et al.</i> , 2010
IFU-fit	Stent-graft suitability vs. anatomy	Device Instructions-for-Use envelopes

WHY THIS MATTERS

Because the maths lives in tested code, the answer is **reproducible and auditable**. The same inputs always give the same score; a reviewer can read the function; a test fails loudly if a threshold drifts. When the agent reports “RCRI 2, class III”, that number came from `compute_rcri`, not from a model’s guess.

4.1 The IFU-fit model

Choosing a stent-graft is an anatomical fit problem. Each device publishes an *Instructions-for-Use* envelope — minimum neck length, maximum angulation, proximal and iliac diameter ranges. Pulse checks the patient’s measured anatomy against every device, per criterion, and returns a verdict with a 10% borderline band, plus a suggested graft-oversizing note. Ranked results tell the surgeon not just *whether* a device fits, but *why* it does or does not.

🔍 UNDER THE HOOD

For a given patient the engine derives calculator inputs from the chart directly — NEWS2 from the latest vitals, RCRI and GAS from the comorbidity profile and labs — and only reports a score when the inputs are actually present. A missing input is never guessed; the score is simply omitted.

The Agentic Brain

Not a chatbot bolted onto a database — a tool-calling agent that plans, calls the tested calculators, remembers the conversation, and streams its reasoning.

Ask the copilot “what’s this patient’s RCRI, and is he suitable for EVAR?” Behind that one sentence, the agent triages the request, fetches the patient record, runs the risk calculator, ranks the devices, and composes an answer that cites what it used. You watch it happen, step by step.

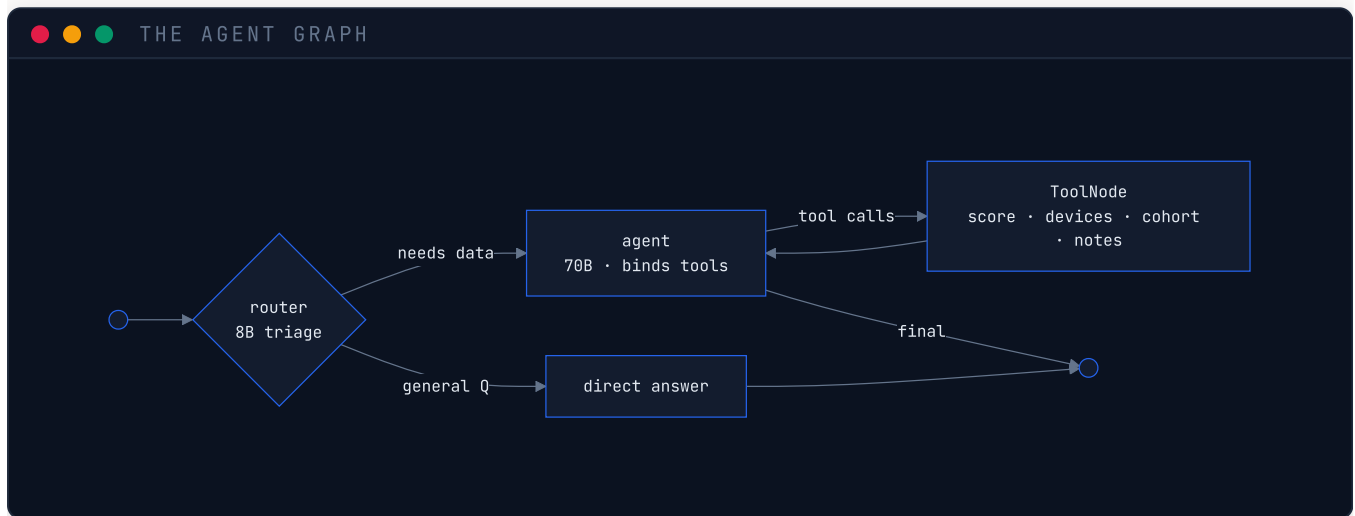


Figure 5.1 – A fast router triages; the agent loops through tools until it has a grounded answer.

5.1 The tools

The agent binds eight grounded tools, each a thin wrapper over code that already exists:

`calculate_risk_score` and `score_patient` (the calculators), `get_patient`, `match_devices` (IFU-fit), `search_guidelines`, `search_device_catalog`, `query_cohort`, and `search_patient_notes`. The system prompt forbids the model from computing a score itself — for a real patient it must call `score_patient`, which derives the inputs from the chart.

5.2 Router, memory, and streaming

A small, fast **router** model decides whether a question needs the tool machinery at all — “what is a Type Ia endoleak?” gets a direct answer; “score patient P001” gets the full agent. Multi-turn **memory** is the conversation itself: prior turns are persisted and replayed, so the graph stays stateless and there is a single source of truth. And every step is **streamed** to the interface as it happens.

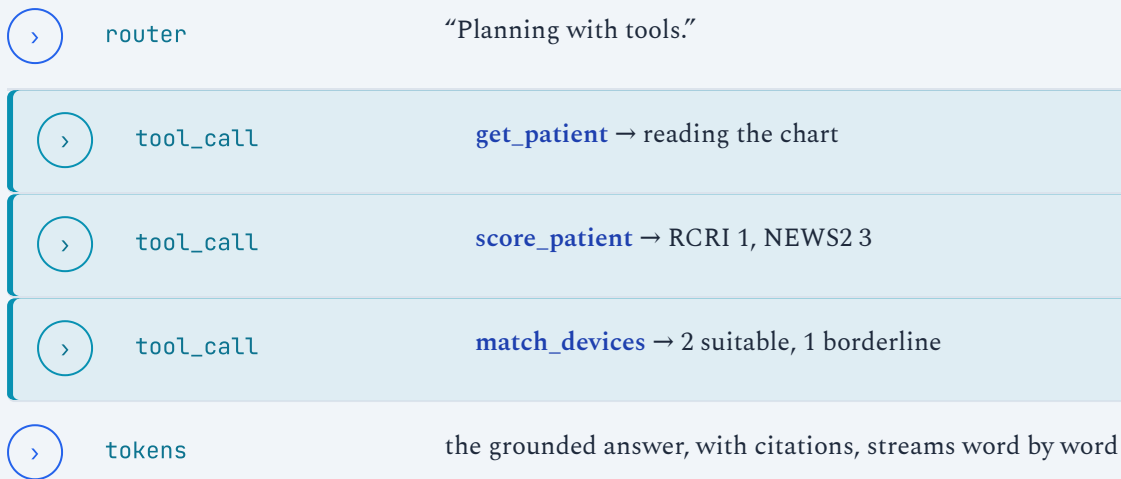


Figure 5.2 – The event stream the copilot renders as a live agent timeline.

GROUNDING, ENFORCED

In live testing the agent once tried to call the calculator with inputs it had guessed. The fix was structural: a `score_patient` tool that derives inputs server-side, so the model *cannot* fudge them. The score computation was always deterministic; now the inputs are too.

Retrieval and Knowledge

When the agent makes a clinical statement, it cites a source. Those sources come from a vector search over a curated knowledge base.

Pulse embeds a small corpus of synthetic guidelines and literature with a sentence-transformer model, and stores the vectors in **Quiver**. When the agent calls `search_guidelines`, the query is embedded, Quiver returns the nearest passages, and an optional cross-encoder reranks them for precision. The passages — title, body, and a citable source — flow back into the answer as source cards.



Figure 6.1 — Embed, search, rerank, cite. Grounding is a pipeline, not a prompt.

🔍 UNDER THE HOOD

Quiver is optional infrastructure: if it is unreachable, retrieval degrades to an empty result set rather than failing the request. The agent then says so honestly and advises consulting local protocols, rather than inventing a citation.

Privacy by Design

The safest patient data is the data you never hold. Pulse holds none — and is built as though it held the most sensitive data there is.

Every patient in Pulse is invented. The ~200 records are generated synthetically — plausible anatomy, plausible comorbidities, plausible vitals — with no connection to any real person. There is, by construction, no PHI to leak. That is an honest mirror of the zero-PHI ethos that serious surgical-AI teams build around.

But absence of data is not an excuse for absence of rigour. Pulse is built with the boundaries a real deployment needs: multi-tenant structure, role-based access on every clinical route, JWT bearer authentication (never a token in a URL), and a fail-closed posture — when the natural-language cohort search cannot map a query, it returns nothing rather than silently matching every patient.

HONEST NON-GOALS

Pulse is a demonstrator, not a certified device. It does not connect to a real EHR, it does not store PHI, and it is not for clinical use. Saying so plainly — on every clinical surface — is part of the design, not a disclaimer bolted on at the end.

The Decisions Behind the Code

The interesting part of a codebase is not what it does, but why it is shaped that way. Pulse records its choices as ADRs.

Architecture decisions, in brief

DECISION	THE WHY
Deterministic scores	Clinical numbers must be reproducible and auditable — so they live in tested code, never in a prompt.
Tool-calling agent	A ReAct graph over the tested calculators gives reasoning <i>and</i> grounding; the dead single-shot RAG was replaced.
Bearer tokens only	JWTs never appear in a URL — they leak via logs, history, and the <code>Referer</code> header. The PDF download fetches with a header.
Cross-encoder rerank	Bi-encoder recall is fast but coarse; a cross-encoder rerank sharpens the top results the agent actually cites.
MCP shares the tool layer	The external MCP surface reuses the same tools as the in-app agent — one source of truth, no drift.

Each of these is written up in `docs/adr/` with its context, the decision, and its consequences — so the next engineer inherits the reasoning, not just the result.

Beyond the App

Pulse's clinical capabilities are not locked inside its own interface. An external agent can drive them over the Model Context Protocol.

The **MCP server** exposes the same grounded tools — score a patient, match devices, look up a patient, search guidelines, query the cohort — over stdio, so any MCP-capable client (including your own assistant) can operate Pulse. It is a thin adapter over the in-app tool layer, so an external agent gets byte-identical, deterministic behaviour: it can no more invent a score than the in-app agent can.

~200

SYNTHETIC
PATIENTS

7

TESTED
CALCULATORS

6

MCP TOOLS

2

LANGUAGES, BY
DESIGN

THE SHAPE OF IT

One deterministic engine. Two ways to reach it — a copilot for people, an MCP server for agents. Both grounded by the same rule: the model orchestrates, the code computes, and every claim is traceable.

⚠ Educational demo on synthetic data — not for clinical use; not medical advice.

Glossary for the newcomer

AAA

Abdominal aortic aneurysm — a focal dilation of the abdominal aorta.

EVAR / TEVAR

Endovascular (thoracic) aneurysm repair — treating an aneurysm from inside the vessel with a stent-graft.

Stent-graft

A fabric-covered metal scaffold deployed to exclude the aneurysm from blood flow.

IFU

Instructions For Use — a device's manufacturer-defined envelope of suitable anatomy.

Endoleak

Persistent blood flow into the aneurysm sac after repair; a Type Ia is a proximal seal failure.

RCRI

Revised Cardiac Risk Index — peri-operative cardiac risk from six factors.

NEWS2

National Early Warning Score 2 — a vitals-based deterioration score.

Embedding

A vector of numbers capturing the meaning of text, so similar meanings sit close together.

RAG

Retrieval-augmented generation — grounding a model's answer in retrieved source passages.

Agent

A model that can call tools in a loop to accomplish a task, not just emit text.

MCP

Model Context Protocol — a standard for exposing tools to external AI agents.

PHI

Protected Health Information — the sensitive patient data Pulse deliberately never holds.

References and further reading

Clinical references

- R1 **Lee TH, et al.** Derivation and prospective validation of a simple index for prediction of cardiac risk of major noncardiac surgery. *Circulation* 1999;100(10):1043–1049.
- R2 **Royal College of Physicians.** National Early Warning Score (NEWS) 2. London, 2017.
- R3 **Samy AK, Murray G, MacBain G.** Glasgow Aneurysm Score. *Cardiovasc Surg* 1994;2(1):41–44.
- R4 **Nashef SAM, et al.** EuroSCORE II. *Eur J Cardiothorac Surg* 2012;41(4):734–744.
- R5 **Wanhainen A, et al.** ESVS 2024 Clinical Practice Guidelines on the Management of Abdominal Aorto-Iliac Artery Aneurysms.

Technical references

- T1 **Quiver** — the security-first vector database. github.com/achref-soua/quiver
- T2 **Model Context Protocol** — modelcontextprotocol.io
- T3 **Pulse source & ADRs** — github.com/achref-soua/pulse

All clinical content in this guide is illustrative and operates on synthetic data. Nothing herein is medical advice.



"From patient to plan, every number is earned.
The model orchestrates; the code computes;
and no claim goes uncited."

```
pulse@ward:~$ ./read --status=complete ✓
```